Quantification: 543-2
=====================
PRE-VALIDATION: no number here has been checked against manual (ImageJ) counts.
PIXEL UNITS: lengths px, areas px^2, densities px^-1. The images carry no micron calibration.
Statistics are over interior lacunae only (those not touching the frame edge).

Per image
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| measure | value |
|---|---|
| lacuna count | 12 |
| interior lacuna count | 12 |
| frame-edge lacuna count | 0 |

Interior lacunae only: mean, median and sample SD (n = 12)
----------------------------------------------------------

| measure | kind | mean | median | sd | units |
|---|---|---|---|---|---|
| area_px2 | lacuna shape | 2027.92 | 2066 | 537.141 | px^2 |
| major_axis_length_px | lacuna shape | 81.5016 | 86.4661 | 16.9465 | px |
| minor_axis_length_px | lacuna shape | 33.8846 | 32.3799 | 6.9212 | px |
| aspect_ratio | lacuna shape | 2.4727 | 2.3986 | 0.6666 | unitless |
| eccentricity | lacuna shape | 0.8942 | 0.9087 | 0.0558 | unitless |
| solidity | lacuna shape | 0.8887 | 0.9115 | 0.0668 | unitless |
| roots_count | headline | 8.5 | 8.5 | 2.7798 | count |
| ring_length_r30_px | headline | 369.583 | 363.5 | 87.4252 | px |
| ring_length_r60_px | reported | 1061.58 | 1079.5 | 212.423 | px |
| owned_length_px | ownership-dependent | 3482.71 | 3215.75 | 1346.8 | px |
| edge_count | ownership-dependent | 161 | 150 | 63.6025 | count |
| mean_edge_length_px | ownership-dependent | 21.6333 | 21.4151 | 2.0651 | px |
| perimeter_px | normalised | 222.101 | 229.693 | 45.067 | px |
| ring_area_r30_px2 | normalised | 8742.17 | 8905.5 | 1127.78 | px^2 |
| ring_area_r60_px2 | normalised | 22189.9 | 23189 | 2825.36 | px^2 |
| in_frame_fraction_r30 | normalised | 0.9965 | 1 | 0.0123 | unitless |
| in_frame_fraction_r60 | normalised | 0.9814 | 1 | 0.0596 | unitless |
| ring_density_r30 | normalised | 0.0419031 | 0.0416567 | 0.00587647 | px^-1 |
| ring_density_r60 | normalised | 0.047547 | 0.0480831 | 0.00496234 | px^-1 |
| roots_per_100px_perimeter | normalised | 3.7455 | 3.6373 | 0.6692 | per 100 px |
| ring_attached_length_r30_px | network v2 | 326.75 | 323.5 | 91.0236 | px |
| ring_attached_length_r60_px | network v2 | 931.083 | 929.5 | 231.731 | px |
| ring_length_w_r30_px | network v2 | 417.216 | 409.099 | 101.117 | px |
| ring_length_w_r60_px | network v2 | 1213.1 | 1229.3 | 244.041 | px |
| sholl_crossings_r10 | network v2 | 9.75 | 9.5 | 2.9886 | count |
| sholl_crossings_r20 | network v2 | 12.8333 | 12 | 2.7907 | count |
| sholl_crossings_r30 | network v2 | 14.3333 | 14.5 | 3.7009 | count |
| loop_count | loops, report only | 19.0833 | 17 | 9.6621 | count |
| loop_edge_share | loops, report only | 0.438 | 0.4396 | 0.0984 | share |
| ring_width_mean_r30_px | width | 7.701 | 7.7684 | 0.3741 | px |

Per field
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| measure | value | units |
|---|---|---|
| analysed_area_px2 | 1.02424e+06 | px^2 |
| total_skeleton_length_px | 44105 | px |
| canalicular_length_density_per_px | 0.0430611 | px^-1 |
| junction_count | 1118 | count |
| junction_density_per_px2 | 0.00109154 | px^-2 |
| lacuna_count | 12 | count |

| | | |
|---|---|---|
| lacunae_per_px2 | 1.172e-05 | px^-2 |
| skeleton_component_count | 150 | count |
| mean_component_length_px | 294.033 | px |
| median_component_length_px | 20 | px |
| field_length_density_w_per_px | 0.04904 | px^-1 |
| field_density_without_flagged_per_px | 0.0430611 | px^-1 |
| field_density_in_roi_per_px | | px^-1 |
| unattached_length_px | 465 | px |
| unattached_component_count | 26 | count |
| canalicular_length_density_attached_per_px | 0.0430611 | px^-1 |
| field_length_density_with_unattached_per_px | 0.0435152 | px^-1 |
| loop_count | 275 | count |
| loop_edge_share | 0.4908 | share |
| edge_count | 2231 | count |

Canalicular width, the new measure

| measure | value | units |
|---|---|---|
| width_mean_px | 8.024 | px |
| width_median_px | 8 | px |
| width_p10_px | 5.6569 | px |
| width_p90_px | 10.198 | px |
| width_pixels_used | 44105 | count |

Canalicular width: 2 x the Euclidean distance transform of the canalicular mask at each skeleton pixel. Measured only on skeleton pixels on the canalicular mask: the lacuna buffer and the dilated vascular mask are excluded. The threads are about 2 to 6 px wide, so the width is close to the pixel grid and quantised in steps of about 1 px, it moves with the threshold (a lower cut widens every thread), and it has not been validated against any manual measurement.

Lacuna shape

| lacuna_id | area_px2 | major_axis_length_px | minor_axis_length_px | aspect_ratio | eccentricity | solidity | orientation_rad | centroid_row_px | centroid_col_px | on_border |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 2494 | 90.6112 | 36.7713 | 2.4642 | 0.914 | 0.9096 | 0.4535 | 88.1403 | 271.181 | False |
| 2 | 3004 | 86.0791 | 52.8366 | 1.6292 | 0.7895 | 0.772 | 0.3181 | 207.791 | 919.033 | False |
| 3 | 1698 | 63.4601 | 34.3535 | 1.8473 | 0.8408 | 0.955 | 0.5365 | 257.425 | 340.805 | False |
| 4 | 1631 | 87.5017 | 30.956 | 2.8266 | 0.9353 | 0.741 | 0.0248 | 297.032 | 38.7695 | False |
| 5 | 1204 | 55.3793 | 28.6478 | 1.9331 | 0.8558 | 0.9177 | 0.1008 | 394.263 | 364.613 | False |
| 6 | 2066 | 81.299 | 34.8468 | 2.333 | 0.9035 | 0.8619 | 0.1545 | 472.247 | 745.532 | False |
| 7 | 1233 | 57.0332 | 27.8479 | 2.048 | 0.8727 | 0.9278 | 0.052 | 459.387 | 484.83 | False |
| 8 | 2066 | 100.448 | 26.9826 | 3.7227 | 0.9632 | 0.9018 | 0.236 | 550.262 | 877.61 | False |
| 9 | 2656 | 108.838 | 32.5449 | 3.3442 | 0.9542 | 0.8877 | -0.0333 | 642.811 | 193.327 | False |
| 10 | 2113 | 91.9731 | 30.2288 | 3.0426 | 0.9444 | 0.9135 | 0.0369 | 728.295 | 633.991 | False |
| 11 | 2173 | 86.8531 | 32.2149 | 2.6961 | 0.9287 | 0.9358 | 0.0768 | 787.166 | 480.067 | False |
| 12 | 1997 | 68.5432 | 38.3841 | 1.7857 | 0.8285 | 0.9402 | 0.1037 | 859.482 | 288.042 | False |

Network per lacuna

| lacuna_id | roots_count | ring_length_r30_px | ring_length_r60_px | owned_length_px | edge_count | mean_edge_length_px | ring_width_mean_r30_px |
|---|---|---|---|---|---|---|---|
| 1 | 12 | 449 | 1329 | 4276.31 | 203 | 21.0656 | 7.9755 |
| 2 | 11 | 561 | 1428 | 5687.42 | 273 | 20.833 | 7.2256 |
| 3 | 6 | 338 | 925 | 2053.68 | 89 | 23.0751 | 7.8081 |
| 4 | 9 | 381 | 709 | 1205.83 | 70 | 17.2262 | 7.1387 |
| 5 | 5 | 279 | 853 | 3453.51 | 161 | 21.4504 | 8.0304 |
| 6 | 10 | 354 | 1023 | 2330.4 | 109 | 21.3798 | 7.4527 |
| 7 | 4 | 219 | 842 | 2977.99 | 113 | 26.3539 | 8.156 |

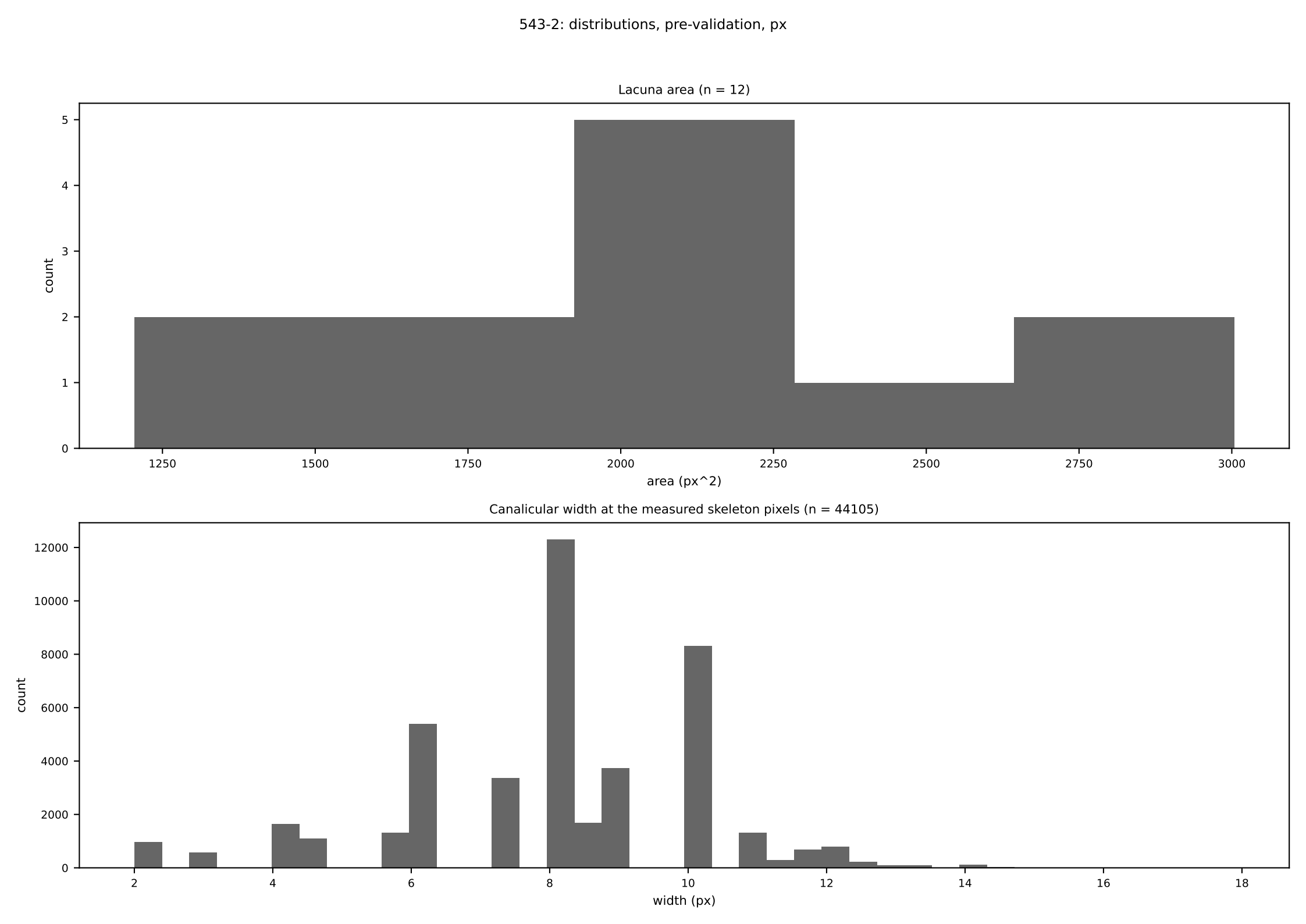
| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 8 | 12 | 425 | 1180 | 2941.22 | 135 | 21.7868 | 7.8171 |
| 9 | 11 | 373 | 1172 | 4403.73 | 201 | 21.9091 | 7.7288 |
| 10 | 8 | 333 | 1136 | 5377.4 | 252 | 21.3389 | 8.2704 |
| 11 | 8 | 408 | 1177 | 4218.45 | 187 | 22.5585 | 7.5027 |
| 12 | 6 | 315 | 965 | 2866.53 | 139 | 20.6225 | 7.3062 |

Normalised per lacuna

| lacuna_id | perimeter_px | ring_area_r30_px2 | ring_area_r60_px2 | in_frame_fraction_r30 | in_frame_fraction_r60 | ring_density_r30 | ring_density_r60 | roots_per_100px_perimeter |
|---|---|---|---|---|---|---|---|---|
| 1 | 244.492 | 9510 | 23791 | 1 | 0.9843 | 0.0472135 | 0.0558615 | 4.9081 |
| 2 | 295.12 | 10519 | 26271 | 1 | 1 | 0.0533321 | 0.0543565 | 3.7273 |
| 3 | 172.894 | 7548 | 18977 | 1 | 1 | 0.0447801 | 0.0487432 | 3.4703 |
| 4 | 232.551 | 8756 | 18925 | 0.9575 | 0.7928 | 0.043513 | 0.0374637 | 3.8701 |
| 5 | 154.267 | 7010 | 17818 | 1 | 1 | 0.0398003 | 0.0478729 | 3.2411 |
| 6 | 231.764 | 9080 | 22988 | 1 | 1 | 0.0389868 | 0.0445015 | 4.3147 |
| 7 | 155.924 | 7017 | 19168 | 1 | 1 | 0.0312099 | 0.0439274 | 2.5654 |
| 8 | 254.25 | 9429 | 23720 | 1 | 1 | 0.0450737 | 0.0497471 | 4.7198 |
| 9 | 277.078 | 10094 | 25807 | 1 | 1 | 0.0369526 | 0.045414 | 3.97 |
| 10 | 227.622 | 8939 | 23523 | 1 | 1 | 0.0372525 | 0.0482932 | 3.5146 |
| 11 | 225.522 | 8872 | 23390 | 1 | 1 | 0.0459874 | 0.0503207 | 3.5473 |
| 12 | 193.723 | 8132 | 21901 | 1 | 1 | 0.0387359 | 0.0440619 | 3.0972 |

Network v2 per lacuna

| lacuna_id | ring_attached_length_r30_px | ring_attached_length_r60_px | ring_length_w_r30_px | ring_length_w_r60_px | sholl_crossings_r10 | sholl_crossings_r20 | sholl_crossings_r30 |
|---|---|---|---|---|---|---|---|
| 1 | 420 | 1261 | 511.546 | 1532.05 | 12 | 16 | 16 |
| 2 | 506 | 1323 | 636.801 | 1628.47 | 15 | 19 | 20 |
| 3 | 294 | 752 | 381.463 | 1062.48 | 9 | 12 | 14 |
| 4 | 355 | 646 | 428.534 | 792.955 | 11 | 13 | 8 |
| 5 | 200 | 589 | 308.48 | 987.749 | 5 | 10 | 12 |
| 6 | 335 | 1001 | 403.706 | 1172.76 | 10 | 11 | 11 |
| 7 | 191 | 749 | 245.752 | 961.495 | 6 | 9 | 10 |
| 8 | 408 | 1032 | 489.304 | 1348.5 | 13 | 12 | 17 |
| 9 | 297 | 837 | 414.492 | 1346 | 8 | 15 | 14 |
| 10 | 252 | 1084 | 377.735 | 1285.83 | 7 | 11 | 15 |
| 11 | 351 | 1041 | 456.605 | 1330.26 | 12 | 14 | 20 |
| 12 | 312 | 858 | 352.179 | 1108.62 | 9 | 12 | 15 |


543-2: distributions, pre-validation, px
Lacuna area (n = 12)
count
area (px^2)
0
1
2
3
4
5
1250
1500
1750
2000
2250
2500
2750
3000
Canalicular width at the measured skeleton pixels (n = 44105)
count
width (px)
0
2000
4000
6000
8000
10000
12000
2
4
6
8
10
12
14
16
18